Derivation appendix: 06 — decays and LFV rates in the soft-broken vacuum

decays and LFV rates in the soft-broken vacuum

1. Soft-vacuum couplings from numeric tensors: $g_{hH^+H^-} = -T \cdot R_{\text{even}} R_C R_C$, $g_{ZAh} = \frac{g}{2c_W}(R_{\text{odd}}^{\mathsf{T}} R_{\text{even}})$, $g_{WH^\pm h} = \frac{g}{2}(R_C^{\mathsf{T}} R_{\text{even}})$

derived here — §2

Independent checks

- all 27 $h_k H_a^+ H_b^-$ agree with boson_vertices to 10^{-9} GeV
- all 9 $ZA_a h_k$ and 9 $WH_a^\pm h_k$ VSS couplings agree to 10^{-12}
- the closed-form VSS width equals the engine's to 10^{-10}

Would be falsified by: any disagreement with the extractor at a soft (non-aligned) point

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2. Off the alignment the charged-lepton fit ($\mu_5 = \mu_4$) is one cubic per eigenvalue-sign pattern; existence depends on $\cos 3\varphi$ only

derived here — §4

Independent checks

- every branch re-diagonalized to the lepton masses (10^{-7})
- every random-start least-squares solution is a cubic branch (10^{-8})
- the draft branch is found, labelled e, at $\varphi = \pi/3$
- the allowed μ_3 set is θ-independent

Would be falsified by: a least-squares solution the cubic misses

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3. The coupling of the vacuum direction is $\text{diag}(m_\ell)/v$ at any vacuum, so SM-like LFV comes only from its $1 - \kappa_V^2$ admixture

derived here — §5

Claim

$$Q(\hat{v}) = \frac{1}{v} \sum_k v_k \, O^{\mathsf{T}} G_k O = \frac{\text{diag}(m_\ell)}{v}$$

Independent checks

- $|Q(\hat{v})| = \text{diag}(m)/v$ to 10^{-14} at 203 soft vacua
- over the scan, the total SM-like LFV rate correlates with $1 - \kappa_V^2$

Would be falsified by: an off-diagonal entry of $Q(\hat{v})$ at any vacuum

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4. Exact-S_3 limit and S_3 image of the soft-vacuum decay machinery

derived here — §3

Independent checks

- at $\varphi = \pi/3$, zero soft terms: the gauge-phobic state, the hH^+H^- trilinears and the LFV couplings reduce to notebook 03's
- at 40 points, $R_{\gamma\gamma}$, VSS widths and all LFV couplings are identical at the S_3 image

Would be falsified by: any observable differing between a point and its image

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5. The LFV width, the diphoton form factors, the CMS limits

imported — §5–6 ([CMS21], [CMS23], [ChakrabartyEtAl21])

Independent checks

- form factors pinned in the main suite (notebook 03, section 6)
- $h \to \gamma\gamma$ decouples to κ_V^2 for heavy $H^\pm$
